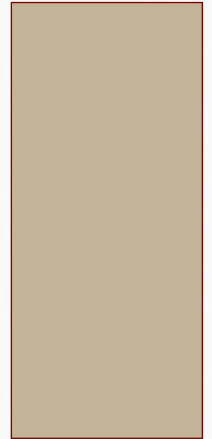


2D VS 3D GAME DESIGN

ADVANCED GAME DESIGN & DEVELOPMENT

DANNY JUGAN



THE THREE PILLARS

- This class will be taught with three modules in mind
- **How-to Generic** – A overview of game design topics in a general, non-platform/SDK manner
- **How-to Specific** – Specific instructions on how to apply those topics in Unity.
- **Design Principles** – Knowing *how* to make games doesn't mean you're going to make a good game. Here we focus on *why* we make certain decisions.

BENEFITS & DISADVANTAGES

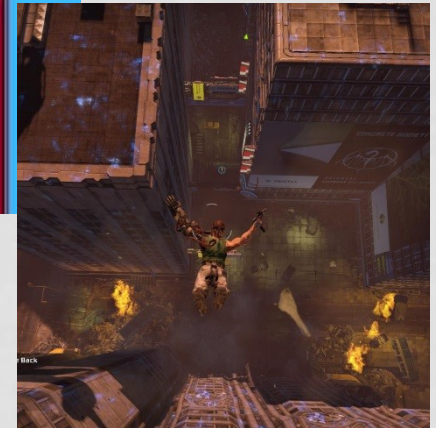
- Compare 2D games with 3D games:
 - 1 – Would you rather play a 3D game or a 2D game?
 - 2 – What advantages do 2D games have?
 - 3 – What advantages do 3D games have?
 - 4 – What changes from the users' perspective/expectation when a game is made in 3D?

WHAT'S THE SAME?

- What game design principles are the same regardless of whether you're designing in 2D or 3D?
- What makes games fun?
 - Easier or harder in 3D?
- How do you engage your players?
 - Easier or harder in 3D?

GOING FROM 2D TO 3D

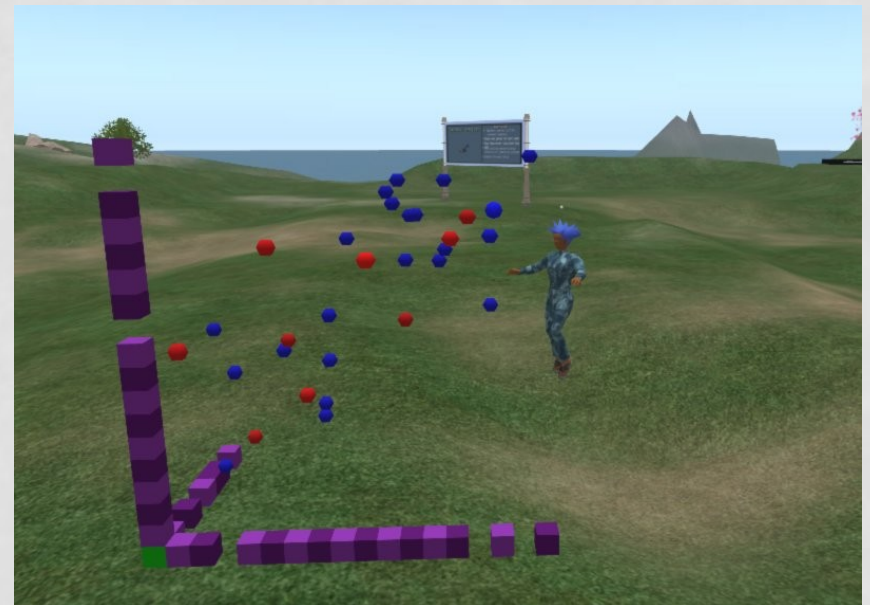
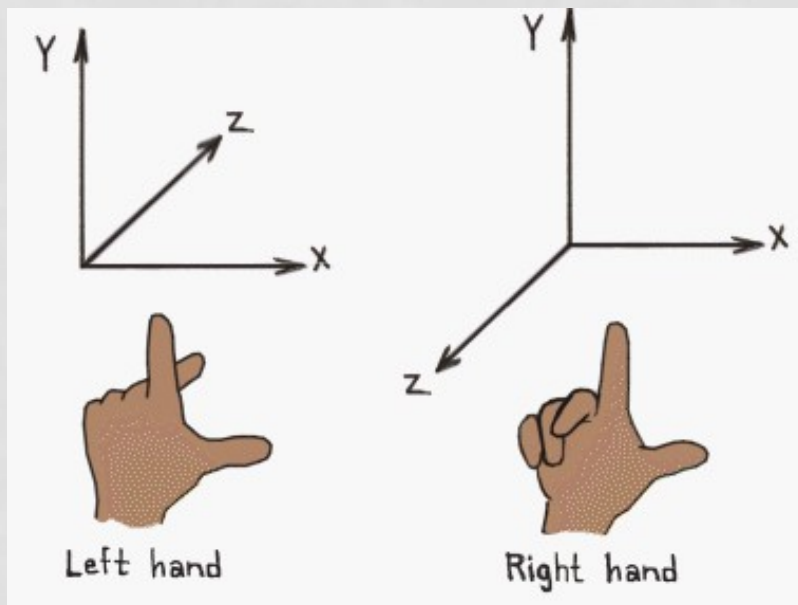
- Can you name some game franchises that started in 2D and went to 3D?



- Were they all successful? Why?

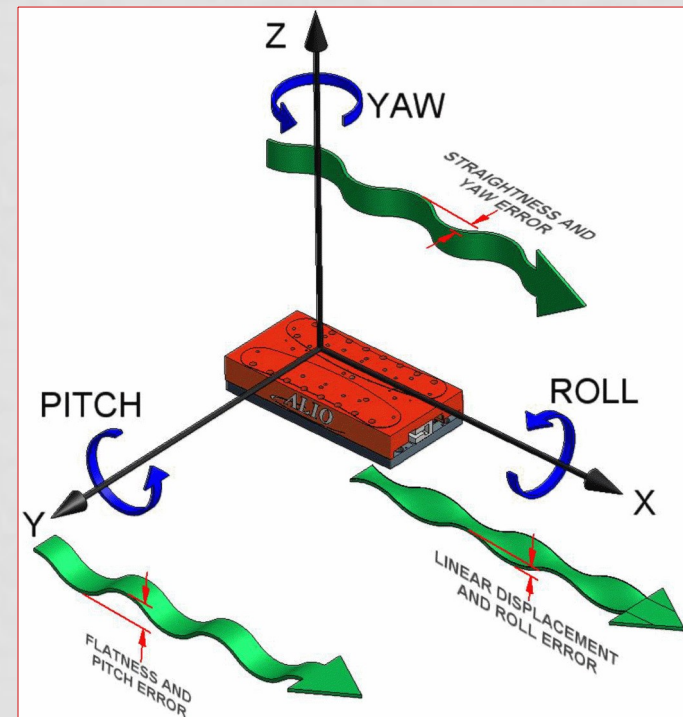
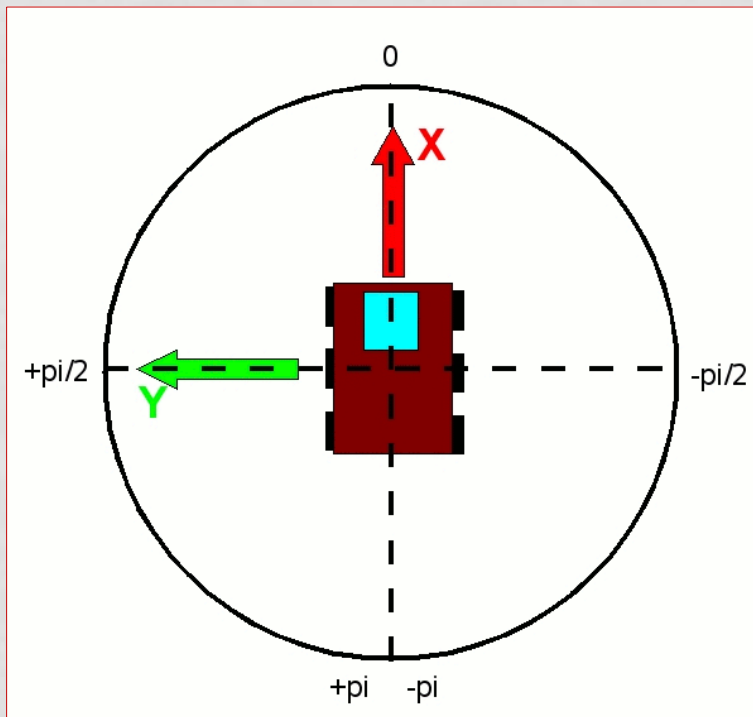
WORLD REPRESENTATIONS

- 2D – (x, y) coordinates
- 3D – (x, y, z) coordinates
 - Unity uses the Left hand system
 - Points in space are represented as Vector3s.



2D ROTATION VS 3D ROTATION

- 2D – Single float value from 0 – 360
- 3D – Euler Angles and Quaternions



CAMERAS

- 3D games employ the use of cameras.
- Cameras determine what the user can see, but not the entirety of what is happening in the game.



CHARACTERS

- 2D
 - Sprite Sheets
 - Vector Graphics
- 3D
 - Models
 - Textures
 - Materials

